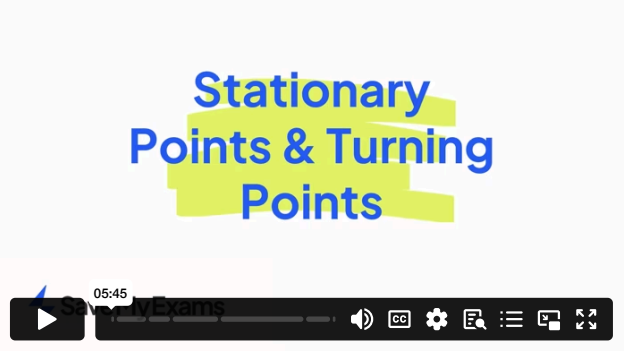
Stationary Points & Turning Points

Exam code: 9MA0

Download PDF

Written by: Roger B
Updated on 20 June 2025

Help me understand



Did this video help you? Yes No

Stationary points & turning points

What are stationary points?

- A stationary point is any point on a curve where the gradient is zero
- To find stationary points of a function **f(x)**

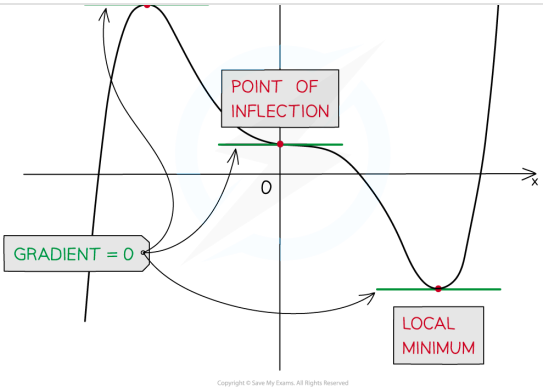
Step 1: Find the first derivative **f'(x)**

Step 2: Solve **f'(x) = 0** to find the *x*-coordinates of the stationary points

Step 3: Substitute those *x*-coordinates into **f(x)** to find the corresponding *y*-coordinates

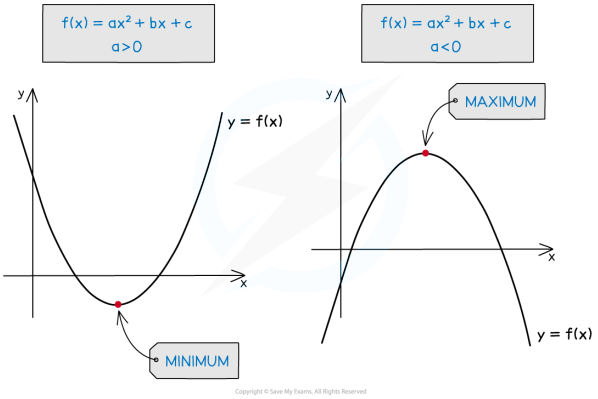
- A stationary point may be either a **local minimum**, a **local maximum**, or a **point of inflection**
- **Turning points** are maximum points or minimum points only
  - The graph turns from going up to down (or down to up)

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



How do I determine the type of stationary point on a quadratic?

- The graph of a quadratic function (ie a parabola) only has a single stationary point
- For an 'up' parabola this is the **minimum**; for a 'down' parabola it is the **maximum** (no need to talk about 'local' here)



- The ***y*** value of the stationary point is thus the minimum or maximum value of the quadratic function
- Quadratics can only have minimum and maximum points
  - so quadratics have **turning points** (also called a **vertex**)

How do I determine the type of stationary point on a curve?

- For a function **f(*x*)** there are two ways to determine the nature of its stationary points
- **Method A:** Compare the signs of the **first derivative** (positive or negative) a little bit **to either side of the stationary point**
  - (After completing **Steps 1 – 3 above** to find the stationary points)

Step 4: For each stationary point find the values of the first derivative a little bit 'to the left' (ie slightly smaller *x* value) and a little bit 'to the right' (slightly larger *x* value) of the stationary point

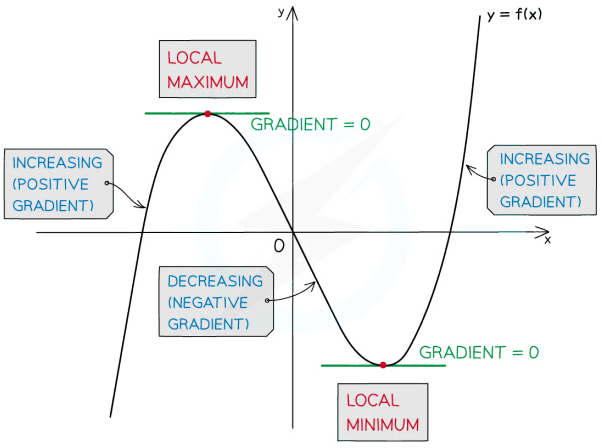
USUALLY WORK HERE (SEE THE EXAMPLE AT THE END)

- DO CHOOSE A CONVENIENT POINT WHERE POSSIBLE ($x = 0$ AND $x = 1$ ARE BOTH VERY EASY TO PUT INTO FORMULAS FOR EXAMPLE)
- DO NOT HOWEVER CHOOSE AN x VALUE TO THE RIGHT OR LEFT THAT JUMPS PAST THE x-COORDINATE OF ANOTHER STATIONARY POINT!

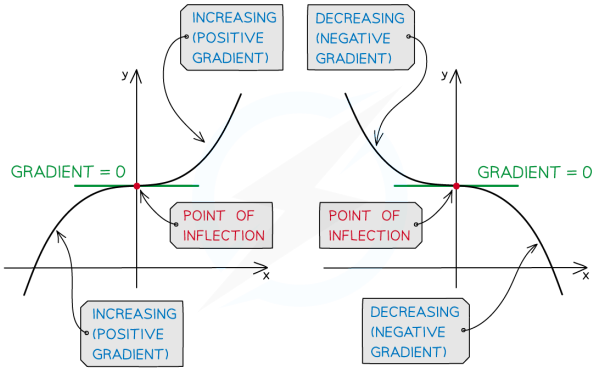
Copyright © Save My Exams. All Rights Reserved

Step 5: Compare the **signs** (positive or negative) of the derivatives on the left and right of the stationary point

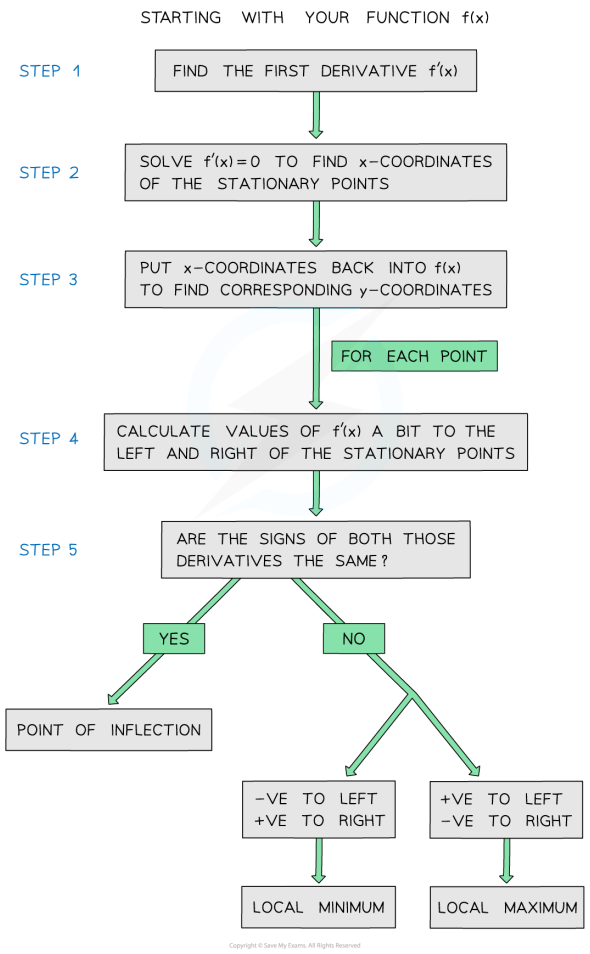
- If the derivatives are **negative** on the **left** and **positive** on the **right**, the point is a **local minimum**
- If the derivatives are **positive** on the **left** and **negative** on the **right**, the point is a **local maximum**
- If the signs of the derivatives are the **same** on both sides (both positive or both negative) then the point is a **point of inflection**



Copyright © Save My Exams. All Rights Reserved



Copyright © Save My Exams. All Rights Reserved



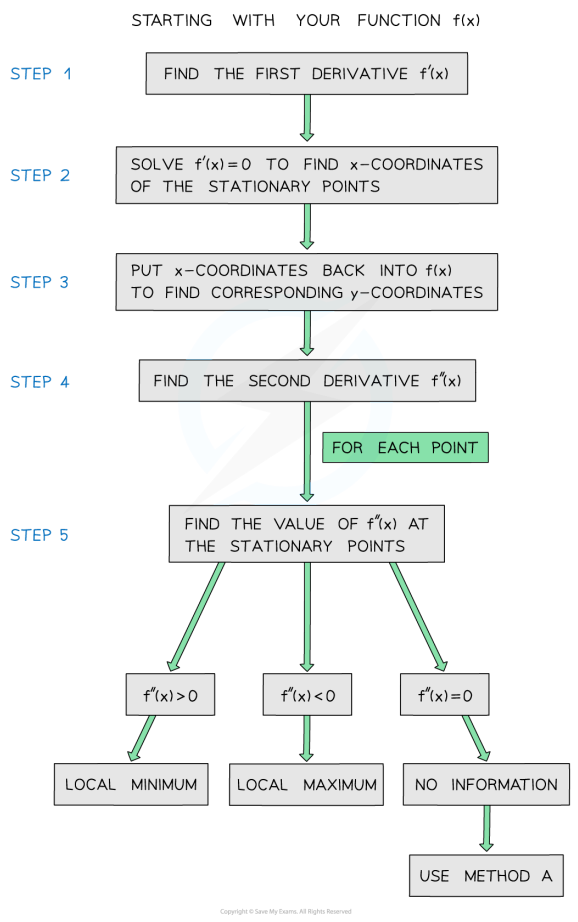
Method B: Look at the sign of the **second derivative** (positive or negative) **at the stationary point**

(After completing **Steps 1 – 3 above** to find the stationary points)

Step 4: Find the second derivative **f''(x)**

Step 5: For each stationary point find the value of **f''(x)** at the stationary point (ie substitute the *x*-coordinate of the stationary point into **f''(x)**)

- If **f''(x)** is **positive** then the point is a **local minimum**
- If **f''(x)** is **negative** then the point is a **local maximum**
- If **f''(x)** is **zero** then the point could be a **local minimum**, a **local maximum** OR a **point of inflection** (use Method A to determine which)



Examiner Tips and Tricks

- Usually using the second derivative (Method B above) is a much quicker way of determining the nature of a stationary point.
- However, if the second derivative is zero it tells you nothing about the point.
  - In that case you will have to use Method A (which **always** works – see the Worked Example).

Worked Example

Teacher only

New! Exclusive Teacher-Only Questions

Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects